

Progress Report

- Increment 1 - Hoo's Fishin'

1) Team Members

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2) Project Title and Description

Hoo's Fishin'

Hoo's Fishin' is a 2D fishing game where the player controls an owl named Hoo and catches fish using a fishing mechanic inspired by Stardew Valley. The player can sell the fish they catch to a shopkeeper and use the money to buy better equipment. As the player continues fishing and earning money, they can improve their gear, catch rarer fish, and unlock more areas of the map.

3) Accomplishments and overall project status during this increment

During this increment, we focused on getting the main gameplay loop into a functional state. We implemented and connected the fishing mechanic, fish spawning, inventory and hotbar system, shop, and money system so the player can catch fish, store them, sell them, and earn money. Some parts were kept simple for testing and functionality and are expected to be improved or changed in later increments. Compared to our original proposal, the main fishing loop is now working in a basic form. The core idea of catching fish, selling them, and using the money as part of progression is in place. However, the project is still missing some of the larger progression features from the original scope, including more upgrades and additional areas of the map. Overall, the project now has a functional foundation that gives us working systems to build on and expand in future increments.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

During this increment, one of the main challenges was learning how to use Godot and becoming more comfortable working with the engine. Most of us had little to no experience with Godot, and it took time to understand how scenes, scripts, nodes, and other parts of the project were organized. We shared tutorials and resources in our group chat to help us understand the engine better, and it helped out a lot.

Another challenge was combining the different game systems into one working project. The fishing, inventory, hotbar, shop, and money systems were developed separately, so connecting them caused smaller bugs and unexpected behavior. We handled these issues by testing the systems together, fixing problems as they appeared, and retesting after changes were made. Since the systems were developed separately, we had to make sure that changes made by different team members did not conflict with each other. As new code was added, we sometimes had to adjust our own work to fit the updated code. Nothing caused a major setback

during this increment, but there were smaller bugs and integration issues that had to be fixed through testing and code reviews.

Our original plan also changed slightly during the increment. Instead of trying to fully develop every planned feature right away, we focused on getting the core systems into a stable and usable state first. This meant that some planned progression and content were moved to later increments so we could build them on top of a working foundation.

5) Team Member Contribution for this increment

Juan Pablo Pastrana II:

- a) Wrote the Stakeholder Communication email section in the progress report document.
- b) Wrote Non-functional Requirements section of requirements and design document.
- c) I did not contribute to the implementation and testing document in this iteration.
- d) Set up the repo and the godot project. Made the MainGame node structure and filled with organized containers. Coded player movement. Made the SceneLoader script that is autoloaded to the project so scenes can be spawned dynamically in code and registers containers to spawn scenes in. Fish load from code instead of being handplaced. Made a fish spawn manager that spawns every type of fish over time on a random side. Made it so Fish despawn overtime when out of view of the player so that fish do not accumulate. Added a shop keeper scene that is interactable when close to the player. When interacting with the shop I made the buy / sell buttons interact with the player wallet. Made the player wallet amount visible as UI. Implemented fish selling by getting data from the players' selected fish in the hotbar and adding that to the balance and removing the selected fish from the inventory. Implemented an upgrade rod button instead of buy button to test player spending money. When the player chooses to upgrade from the shop UI it changes the player's rod data to the next level. Made a pause menu that pauses the game when the user presses the "esc" key. The pause menu resumes the game or exits the game based on what the user selects from the pause menu. Finally, I reviewed and merged my teammates' PRs.
- e) I demonstrated the game demo for the video presentation.

Tyler Dussault:

- a) I was not responsible for updating the progress report for this increment. I'll be in charge of it next increment.
- b) I helped list a lot of the features expected in the requirements, helping us get started. I also wrote the description for what the game is about, allowing my team members to visualize my game idea.
- c) I was not responsible for writing anything in the implementation and testing document.
- d) For the source code, I added the background assets and applied a parallaxlayer to each of the 8 pieces. The closer pieces of the background were made to move faster (the water) while the further pieces moved slower (the sky). I also hand-created the assets for the fishing minigame and added the code required to initiate the fishing game and catch the fish.
- e) I am not responsible for the video in this increment. I'll be in the video for the next one.

Bridget Gizzi:

- a) I was not responsible for contributing to the progress report in this increment but will be responsible for sections in increment 2.
- b) I wrote the Operating Environment and Assumptions and Dependencies for the requirements and design document.
- c) I was not responsible for contributing to the implementation and testing in this increment but will be responsible in increment 2.

- d) I was responsible for creating the owl character's sprite animations. I set up an AnimatedSprite2D node in Godot and designed 32x32 pixel art for the owl's idle and flying poses. I then wrote the GDScript logic in hoo.gd to switch between these animations based on the player's movement input, including flipping the sprite's direction depending on which way the owl was moving.
- e) I described our change in scope and our plan for the next increment in the video presentation.

Darius Davis:

- a) I was not responsible for writing a section of the progress report during this increment, but I contributed updates about the features I implemented and tested for the team.
- b) I contributed to the design and development of the fish progression, fishing rod, and shop purchasing systems. I helped define different fish rarity levels and how different fishing rods affect which fish the player can catch. I also contributed to the design of the fish visuals and swimming behavior.
- c) I tested the fish, fishing rod, shop purchasing, and swimming systems throughout development. I tested rod compatibility with different fish rarities, catch-speed differences between rods, successful and unsuccessful shop purchases, insufficient funds, duplicate rod purchases, player wallet updates, fish movement, swimming boundaries, sprite direction changes, and interactions between the player and swimming fish.
- d) For the source code, I implemented the fish rarity and fishing rod progression system. I created fish resources for Bluegill, Bass, and Golden Trout with different rarity levels, sell values, and catch difficulty values. I implemented Basic, Improved, and Advanced fishing rods with different fish-rarity limits and catch-speed multipliers. I also worked on the shop purchasing backend by creating the player wallet and purchasing logic. The system checks whether the player has enough money, deducts the correct amount, tracks purchased rods, prevents duplicate purchases, and provides a purchase signal for integration with the shop system. I also implemented the visual swimming-fish system. I created pixel-art visuals for the Bluegill, Bass, and Golden Trout and created a reusable SwimmingFish scene and GDScript movement system. The fish swim at different speeds, gently move vertically, randomly change direction, automatically flip their sprites when turning, and remain within configured water boundaries. I configured the fish so they do not physically block the player while swimming. I also connected each swimming fish to its corresponding existing Fish resource so its visual representation uses the same fish data as the fishing system.
- e) I am not responsible for the video in this increment, but i will be in the video in the next increments.

Dhruv Patel:

- a) Wrote the accomplishments and overall project status section, including the progress made on the main gameplay loop and how the project compares to the original proposal. Also wrote the challenges and changes section, including learning Godot, connecting separate game systems, fixing issues through testing, and keeping some systems simple for current functionality.
- b) Worked on the inventory and hotbar sections. Documented how items are stored in the inventory, how item stacking works, how items are displayed in the hotbar, how the player selects hotbar slots, and how the hotbar connects to the inventory system. Worked on the diagrams for the systems implemented.
- c) Wrote the programming language, technologies used, functional testing, non-functional testing, and non-execution-based testing sections. Documented the use of GDScript and Godot, how gameplay features are tested by running the game, and how GitHub, group chat discussions, meetings, code reviews, and walkthroughs are used to review changes.
- d) Worked on the inventory and hotbar parts of the game. Implemented inventory storage and item stacking so multiple copies of the same fish can share a slot while keeping track of the quantity. Worked on the hotbar so inventory items and their quantities are displayed in the correct slots and the player can switch

between them. Added a selected-item label above the hotbar that displays the name of the currently selected fish or fishing rod and remains blank when an empty slot is selected. Also worked on connecting the hotbar to the inventory so changes to stored items are reflected in the hotbar.

- e) I went over the general overview of the project and gave a description of what was accomplished during this increment and state of our project in the video.

6) Plans for the next increment

- Functional main menu screen
- Improved fishing minigame (more polish)
- More fish types
- More fishing rods and equipment
- Bait and lure items
- Expanded shop and upgrade options
- Player tasks based on fish type, weight, or other goals
- Backpack upgrade for increased inventory space
- Achievenet tracking system

7) Stakeholder Communication

Hoo's Fishin' Progress Report One

Hello, I am emailing to communicate the progress we have made thus far in the development of the game "Hoo's Fishin'" and where we are heading with the project.

The game is in a playable state, the player can walk around, interact with the shop and fish. Fish spawn around the player and swim. Once a fish has been clicked, the fishing minigame begins where the player has to balance the cursor on the fish by pressing space. If the player successfully balances then they are rewarded with the fish that they caught. Next, the player can sell the fish to get money that can then be used to upgrade rods.

Some setbacks that we experienced while developing "Hoo's Fishin'" were having to learn and adopt a new engine. Time was dedicated to experimenting and understanding what we have to work with and a good Godot workflow. Also, systems that we built were built separately and then combined which caused some integration bugs. Every feature was tested together before moving on. Finally, our scope changed because we want to build a game that can be scaled up simply so new features are not difficult. That is why the prototype does not have many variations.

What we are looking forward to developing next for "Hoo's Fishin'" is more fish types, different rods, new baits, lures, and achievements that can help the player progress in the game. Some bug fixing will come along the way and the implementation of a main menu.

Best regards,

The Hoo's Fishin' Team

8) Link to video

 Increment1_Video.mp4